Multimodal Satellite Change Detection for Urban Development and Environmental Monitoring

Manish Y. M.
Polaris School of Technology, Medhavi Skills University
manishym946@gmail.com

Abstract

When the post-change Sentinel-2 image is missing or cloudy, a SAR–optical change detector must work from Sentinel-1 and, at best, an older optical image. We test whether that historical image helps, on the 10 test cities of the OSCD urban change benchmark, with the official dual-stream U-Net (DS-UNet), U-Net baselines, 10 training seeds per model, validation-only selection and a pre-registration written before each test evaluation. In DS-UNet trained with modality dropout, whose optical stream meets SAR only at the last layer, the historical image made no detectable difference (ΔF1 +0.007, 95% CI [−0.004, +0.018]). In an early-fusion U-Net trained with modality dropout, where it can condition the SAR features, it lowered F1 (−0.091; replicated on 10 new seeds, −0.116) by raising false alarms. The same early-fusion model given no optical input scored above a U-Net trained on SAR alone (+0.072 on new seeds), an effect concentrated in one city. The image's acquisition time (1–18 months) showed no detectable linear trend. On real cloudy scenes for 7 cities, a model trained without dropout fell far below SAR alone (−0.275), while the dropout model with cloud pixels masked recovered most of the loss (+0.208) and scored slightly above SAR alone (+0.020); discarding the cloudy image did at least as well as masking it. Results concern one benchmark, few cities and one training recipe.

Keywords: change detection, SAR–optical fusion, Sentinel-1, Sentinel-2, missing modality, modality dropout, historical optical imagery, cloud cover, OSCD, pre-registration

1. Introduction

Optical satellite images carry the spectral detail that makes urban change easy to recognise, but on the date a change must be assessed they are often unusable because clouds cover part or all of the scene. Synthetic aperture radar (SAR) is unaffected by cloud, which motivates SAR–optical fusion for change detection [1]. Most fusion models are trained and evaluated with SAR and optical images at both dates. When the post-change optical image is missing, an older, cloud-free optical image is often available, and Song et al. [2] used one as semantic context for a SAR-based detector of construction-ready bare land, reporting that its usefulness falls as it ages. Methods for a partly missing optical modality either reconstruct the missing features [12] or train with the modality randomly removed [11]. Whether a historical optical image adds change evidence to bitemporal SAR, and how a model trained for missing optical data behaves on real clouds, have not been measured on a public change detection benchmark.

We address five questions:

- **RQ1:** Does historical Sentinel-2 imagery improve Sentinel-1 change detection when the post-change optical image is unavailable?
- **RQ2:** How does the time between the historical image and the post-change date affect it?
- **RQ3:** Does a second polarisation (VH with VV) improve SAR change detection or reduce false alarms?

- **RQ4:** How do the models that work without the post-change optical image compare with SAR-only, optical-only, input-fusion and dual-stream models?
- **RQ5:** Under which temporal and land-cover conditions does the historical image help most?

We answer them in a measurement study on OSCD [3] enriched with Sentinel-1, using the official DS-UNet code [1], not a new method. Our findings are:

- **(i) The historical image does not help, and where it can interact with SAR it hurts.** In DS-UNet with modality dropout, adding it changed F1 by +0.007 (not detectable). Because DS-UNet fuses the streams only at the last layer, we also trained an early-fusion model in which the historical image enters the same encoder as SAR: there it lowered F1 by 0.091 through false alarms, and the loss replicated on new seeds. The null in DS-UNet is therefore not an artefact of late fusion.
- **(ii) Training with optical data and then removing it gave a better SAR-only detector** than training on SAR alone (+0.072, confirmed on new seeds), although the gain is concentrated in one test city.
- **(iii) Missing and cloudy are different failures.** A dropout-trained model handles a missing post-change image but reads unmasked cloud as change, so for it a cloudy image is worse than none; zero-filling the cloud pixels recovers most of the loss, and on real scenes discarding the cloudy image did at least as well as masking it. A model trained only with full input fails in both cases and, on real clouds, falls far below SAR alone.
- **(iv) Acquisition timing and polarisation:** no detectable linear trend over 1–18 months, with a confounded 1-month low; for SAR alone, adding VH to VV lowered F1.

Because OSCD is small and seed variance alone can reverse benchmark rankings [17], every result uses 10 paired training seeds and Holm-corrected tests [4]; every checkpoint and threshold is selected on validation cities only; and a pre-registration with code hashes was written before each test evaluation (committed to version control from Step 4B on). Analyses chosen after results were seen are labelled exploratory.

2. Related Work

SAR–optical change detection. Siamese fully convolutional networks established bitemporal change detection on optical images [5], with U-Net [6] variants as common backbones; reviews [7] cover the field. Hafner et al. [1] fuse Sentinel-1 VV and Sentinel-2 in a dual-stream U-Net with decision-level fusion and report that it beats unimodal U-Nets and input-level fusion on OSCD enriched with Sentinel-1. Ebel et al. [8] fuse co-registered Sentinel-1 and Sentinel-2 in Siamese networks, Saha et al. [9] study self-supervised change detection across sensors, and DSNUNet [10] combines Sentinel-1 and Sentinel-2 for forest change. These methods assume both modalities at the dates compared. Heterogeneous change detection instead compares an optical image at one date with a SAR image at the other, for example by coupling or translating the two image domains [25], [26], and multimodal optical–SAR datasets now support all-weather damage assessment [27]. In our setting SAR is available at both dates and the historical optical image is auxiliary.

Historical optical guidance. Song et al. [2] monitor construction-ready bare land in Guangzhou with Sentinel-1 VH change screening and a historical cloud-free Sentinel-2 image. Varying the image's age (1, 3, 6 and 12 months before the target date) lowered F1 from 83.99% to 78.31%; these gap results report no repeated runs. In their framework the historical image is a semantic prior for land cover at the target date. In our bitemporal setting it fills the pre-change slot, where an image that already shows the change hides it, so the expected sign of a timing effect differs and our results are not a replication.

Missing modalities and clouds. Modality dropout was introduced for gesture recognition [11]. Hafner and Ban [12] reconstruct optical features from SAR for multi-temporal urban mapping with a partly missing

optical modality; on samples with missing optical data their model beats zero-fill (F1 0.327 against 0.233) but falls slightly short of a SAR-only model (0.348). SAR-guided cloud removal reconstructs cloud-free Sentinel-2 images [13], and SEN12MS-CR-TS provides real cloudy Sentinel-1/2 time series [28]. Real cloud masks are available in the Sentinel-2 Cloud Mask Catalogue [14]; operational detectors such as s2cloudless [15] are imperfect, and an intercomparison of cloud-masking algorithms documents large differences between methods [16]. Masking detected cloud and treating it as missing data is standard practice; we measure how its errors interact with a dropout-trained change detector.

Evaluation practice. Variance from data sampling, initialisation and hyperparameters can change benchmark conclusions [17]; Appendix A shows a ranking reversal on OSCD under a weaker protocol.

3. Data and Methods

3.1 Data and split

OSCD. OSCD [3] provides 24 bitemporal Sentinel-2 L1C image pairs with pixel-level urban change labels. We use the 13 bands, divided by 10000 and clipped to [0, 1]. Ten of the 14 official training cities are used for training, the other 4 (aguasclaras, beihai, bercy, paris) for validation, and the 10 official test cities (brasilia, montpellier, norcia, rio, saclay_w, valencia, dubai, lasvegas, milano, chongqing) for testing. Test cities never enter training or validation, which is asserted in code. Change prevalence on the test cities is 0.052; labelling every pixel as change gives F1 0.098.

Sentinel-1. VV rasters released with DS-UNet [1] (3-month means before date 1 and after date 2), resized to the label grid by cubic interpolation; training samples an orbit at random and evaluation uses the first listed orbit. For RQ3 we exported VV and VH from Google Earth Engine [18] with DS-UNet's own recipe (VH range [−30, 0] dB, fixed before download); the exported VV equals the released raster pixel for pixel in all 102 files. Las Vegas has no VH at date 1 on its evaluated orbit, so the RQ3 analysis uses 9 test cities. Date-1 windows hold fewer scenes than date-2 windows in 22 of 23 test-city orbits (median 6 against 14 VV scenes), so date-1 means are noisier.

Historical Sentinel-2 images (RQ2). For each city, one Sentinel-2 L1C scene at each target date 2 − g, g ∈ {1, 3, 6, 12, 18} months, and one at OSCD's own date-1 day as a control of the processing chain. Rule, fixed before any model saw a scene: within ±45 days of the target, full coverage required, the day nearest the target with city-box cloud fraction ≤ 0.02, else the clearest day. The primary set is the 9 test cities whose scene has cloud fraction ≤ 0.10 at every gap (Chongqing excluded). Alignment against OSCD's own date-1 bands gives median band correlation 0.940 and median shift 0.27 px. At 1–12 months no primary city's image precedes OSCD date 1; at 18 months, 4 of the 9 do.

Real cloudy scenes (Step 11). For each test city, the full-coverage Sentinel-2 L1C day within ±30 days of OSCD date 2 whose city-box cloud fraction (s2cloudless probability ≥ 40) lies in [0.05, 0.70], nearest to date 2; Montpellier, Rio and Las Vegas had none and contribute their nearest scene with cloud ≥ 0.90 ("overcast"). Each city also has OSCD's own date-2 day as a control. The rule was applied to stored cloud fractions before download; the scenes and their cloud fractions are listed in the Step 11 pre-registration.

Land cover. ESA WorldCover 2020 v100 [19] on the label grid (nearest neighbour). It post-dates OSCD date 2 and is used as a stratification, not as a label; OSCD has no labels separating urban from agricultural or seasonal change.

3.2 Models and training recipe

The network and loss are the official DS-UNet files (`networks.DualStreamUNet`, topology [64, 128, 256, 512]; `jaccard_like_loss`) loaded at run time from commit 8bfa4be [1]. In DS-UNet the SAR and optical streams are separate U-Nets that meet only at a final 1×1 convolution over their concatenated features, so the optical stream never sees SAR. The U-Net baselines are DS-UNet's single-stream U-Net with the same recipe. Training: Adam, learning rate 5e-5, weight decay 5e-4, batch 64, 32×32 importance crops, flips and 90° rotations, 2500 steps, validation every 25 steps. The step budget was chosen on validation score only (390 steps: 0.354; 1000: 0.382; 2500: 0.405). Training runs with cuDNN determinism (PyTorch 2.5.1+cu124, one RTX 4060 laptop GPU). Arms with the same seed share the generator that draws cities, crops, augmentations and input conditions; contrasts between arms are paired by seed. As a reference, the official 390-step protocol on all 14 training cities gives test F1 0.557 ± 0.035, below the published 0.600 [1]; our conclusions concern differences between arms under one shared protocol.

3.3 Arms

A missing optical image is represented by zeros in its channels. Conditions: `full`; `no_o2` (post-change optical zero, pre-change optical image present: the historical-image setting); `no_opt` (both optical dates zero: SAR only).

arm	network	training conditions	step
A	DS-UNet	full input	Phase 2
B	DS-UNet	full 0.50; `no_o2` 0.25; `no_opt` 0.25 (modality dropout)	Phase 2
C	DS-UNet	full 0.50; post-change optical zero under a cloud-shaped mask 0.25; `no_o2` 0.125; `no_opt` 0.125	Step 4B
async	DS-UNet	`no_o2` only	Step 6
SAR-only, optical-only, input fusion	U-Net	full input	Step 6
E	U-Net, VV and optical stacked at the input	as B (modality dropout)	Step 9 (seeds 1–10), Step 10 (seeds 11–20)
VH, VV+VH	U-Net and DS-UNet	full input	Step 7

Each arm is trained with seeds 1–10, and E and the SAR-only U-Net again with seeds 11–20 (Step 10). In `no_o2` the historical image is OSCD's own date-1 image or, for RQ2, one of the historical scenes in the pre-change slot. E is the early-fusion counterpart of B: in E the historical image enters the same encoder as SAR, so it can condition the SAR features.

Selection (validation cities only). Checkpoint: the step with the highest mean validation F1 over the conditions the arm trains on. Threshold: the validation-F1 argmax over {0, 0.01, …, 1} for each input condition, frozen and applied unchanged on test. Where a model is evaluated on inputs it has no threshold for (clouds, §5), its frozen full-input threshold is used.

3.4 Statistics and integrity

F1, precision, recall and false-alarm rate (FP / (FP + TN)) are computed from pixel counts pooled over the analysed test cities. Each step has pre-specified families of paired comparisons over seeds (two-sided paired t-test, Holm correction [4]), reported with the mean difference, its 95% CI, d_z and the number of seeds with a positive difference; a Wilcoxon test is computed alongside. Labels such as "higher" require Holm $p < 0.05$; otherwise "no detectable difference", which is not evidence of equivalence. With 10 pairs the minimum detectable effect at 80% power is about 1.00–1.22 SD of the paired differences, depending on the family.

Intervals describe variation across training seeds on fixed test cities, not across cities; §4.7 examines city variability (exploratory).

Pre-specification. Each step's pre-registration (hypotheses, primary family, decision rule, SHA-256 hashes of code, inputs and checkpoints) was written before its first test evaluation; for Steps 4B–11 it was committed to version control first, and from Step 9 on a gate script refused to run unless the committed file and every hash were unchanged. These are repository files, not public registrations. "Step 4B" is the spatial-dropout arm C of the robustness study; "Stage 4B" is a later, separate evaluation of B with and without the historical image. Evaluations enforce anchors, for example that re-evaluated test counts equal stored ones exactly. Audit scripts written with the same AI assistant recompute the statistics of Phases 2–8 (Appendix D). Analyses added after results were seen are labelled exploratory; one of them (§4.3) was then tested on new seeds as a registered hypothesis.

Figure 1 summarises the registered primary contrasts of the study.

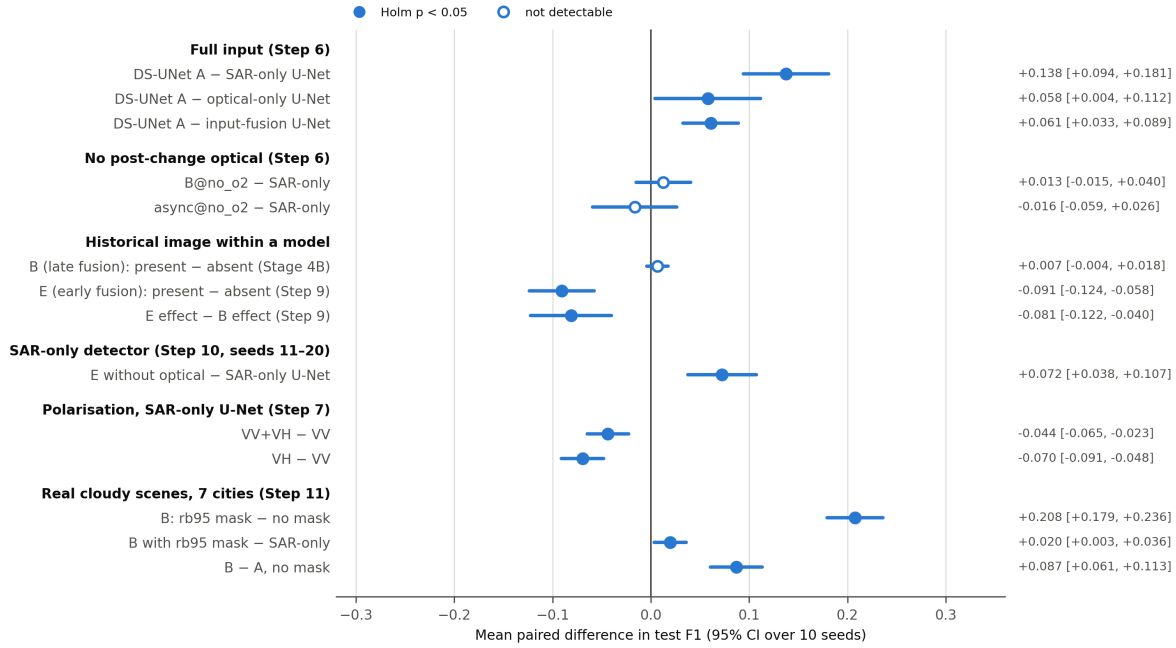


Figure 1. Registered primary paired contrasts (mean difference in test F1 with 95% CI over 10 training seeds; filled marker: Holm $p < 0.05$). Stage 4B is a single test.

4. Results

4.1 RQ4: comparison with SAR-only, optical-only, conventional fusion and dual-stream approaches

Full input, test F1 at the validation-selected threshold, 10 test cities (Step 6; primary family A – each baseline, Holm over 3):

arm	F1	precision	recall	IoU	false-alarm rate
DS-UNet A	0.556 ± 0.029	0.568 ± 0.071	0.562 ± 0.090	0.385 ± 0.028	0.025 ± 0.013
DS-UNet B (modality dropout)	0.553 ± 0.030	0.466 ± 0.041	0.682 ± 0.032	0.382 ± 0.028	0.043 ± 0.007
SAR-only U-Net	0.418 ± 0.043	0.312 ± 0.051	0.648 ± 0.027	0.265 ± 0.035	0.081 ± 0.020
optical-only U-Net	0.498 ± 0.056	0.588 ± 0.080	0.451 ± 0.101	0.333 ± 0.048	0.019 ± 0.012
input-fusion U-Net	0.495 ± 0.026	0.556 ± 0.063	0.452 ± 0.047	0.329 ± 0.023	0.020 ± 0.006

contrast	mean Δ	95% CI	d_z	Δ > 0	Holm p	label
A – SAR-only	+0.138	[+0.094, +0.181]	+2.26	10/10	1.6e-04	DS-UNet better
A – optical-only	+0.058	[+0.004, +0.112]	+0.77	8/10	0.0370	DS-UNet better
A – input fusion	+0.061	[+0.033, +0.089]	+1.54	9/10	0.0017	DS-UNet better

Across training seeds, the dual-stream design scores above all three U-Net baselines, in line with Hafner et al. [1]. Input fusion is not distinguishable from optical alone (Δ −0.003, 95% CI [−0.045, +0.039]). The recipe and step budget were chosen on arm A, the baselines were not tuned separately, and the selected thresholds of the optical-only and input-fusion U-Nets reach 0.99 or above in several seeds, which suggests poorly scaled outputs. In the exploratory city resampling (§4.7), A's lead over SAR-only is concentrated in Las Vegas and not established beyond these cities.

Without the post-change image. B and async without the post-change image are not detectably different from the SAR-only U-Net (+0.013, −0.016; registered, §4.2) and, in an exploratory comparison, score below the optical-only, input-fusion and DS-UNet models that have that image.

4.2 RQ1: the historical image, with late and with early fusion

Against SAR alone (Step 6). Setting `no_o2` against the SAR-only U-Net, 10 test cities (primary family, Holm over 2):

contrast	F1	mean Δ	95% CI	d_z	Δ > 0	Holm p	label
B@no_02 – SAR-only	0.431 ± 0.034	+0.013	[−0.015, +0.040]	+0.33	6/10	0.6576	no detectable difference
async@no_02 – SAR-only	0.402 ± 0.059	−0.016	[−0.059, +0.026]	−0.27	5/10	0.6576	no detectable difference

The F1 column is the first term; the SAR-only U-Net scores 0.418 ± 0.043. Neither model is detectably better than SAR alone. The companion study (Appendix A) had reported a margin for its missing-optical model under a weaker protocol; under the shared protocol it does not survive.

Within the late-fusion model (Stage 4B). At B's frozen full-input threshold, the threshold used for every corrupted input in §5, B without the post-change image scores above the SAR-only U-Net (+0.060). Stage 4B decomposes this margin by evaluating B with the historical image present and absent (both optical slots zero):

Table 4. B at its frozen full-input threshold, 10 test cities (Stage 4B; primary: present – absent, single test; other rows descriptive).

contrast	mean Δ	95% CI	d_z	Δ > 0	p	label
B: image present – absent	+0.007	[−0.004, +0.018]	+0.44	7/10	0.198	no detectable difference
B, image present – SAR-only	+0.060	[+0.026, +0.094]	+1.27	8/10	0.00303	(descriptive)
B, image absent – SAR-only	+0.053	[+0.022, +0.085]	+1.20	9/10	0.00415	(descriptive)

The historical image contributes no detectable F1 (it raises recall by 0.018, Holm p 0.030, with no detectable change in precision). B's margin over the SAR-only U-Net comes from B itself: with no optical input at all it already scores 0.053 above it. At B's own `no_02` threshold the within-B contrast is also not detectable (+0.002). Stage 4B was registered after the within-B contrast had been seen at two other thresholds, which its pre-registration states.

With early fusion (Step 9). DS-UNet's late fusion lets the historical image add only a SAR-independent term to each pixel's score. To test whether that explains the null, we trained E, a U-Net with VV and the 13 optical bands stacked at the input, with B's dropout mixture (validation-selected thresholds per condition, as Step 6):

Table 5. Step 9, E against B and the baselines, 10 test cities (primary family Holm over 2; secondary families Holm over 2 each).

contrast	mean Δ	95% CI	d_z	Δ > 0	Holm p	label
E: image present – absent	−0.091	[−0.124, −0.058]	−1.96	0/10	0.0003	lower with the image
(E present – absent) – (B present – absent)	−0.081	[−0.122, −0.040]	−1.41	2/10	0.0016	early fusion lower
E@no_02 – SAR-only	−0.019	[−0.072, +0.034]	−0.26	5/10	0.4383	no detectable difference
E@no_02 – B@no_02	−0.032	[−0.068, +0.005]	−0.63	2/10	0.1590	no detectable difference
E@full – input fusion	+0.040	[+0.006, +0.074]	+0.85	9/10	0.0496	E higher
E@full – A	−0.021	[−0.055, +0.013]	−0.44	3/10	0.2012	no detectable difference

In E the historical image lowers F1 by 0.091 (0.399 ± 0.039 with it, 0.490 ± 0.010 without it), in every seed, and the difference from its effect in B is detectable. It does so through false alarms (exploratory): precision falls by 0.133 (95% CI [−0.195, −0.071]) and the false-alarm rate rises by 0.041, with no detectable change in recall. This pattern is consistent with the network reading an optical image at date 1 next to zeros at date 2 as change, although dropout training showed it this input in a quarter of its samples; we did not test the mechanism directly. On 10 new seeds (Step 10, Table 6) the loss replicates (−0.116). On full input E is not detectably different from A and scores above the full-input-only input-fusion U-Net (Holm p 0.0496).

Answer to RQ1. On OSCD, historical Sentinel-2 imagery did not improve Sentinel-1 VV change detection without the post-change optical image. In the late-fusion DS-UNet it made no detectable difference (95% CI [−0.004, +0.018] at the full-input threshold); in an early-fusion U-Net, where it can condition the SAR features, it lowered F1 by raising false alarms. The DS-UNet null is therefore not explained by late fusion alone. Effects below about 1.22 SD of the paired differences can be missed, and the Sentinel-1 inputs are 3-month means.

4.3 A dropout-trained network as a SAR-only detector

Table 5's model E, evaluated with both optical slots zero, scored above the SAR-only U-Net on seeds 1–10 (+0.072, 95% CI [+0.045, +0.099], 10/10; exploratory, chosen after the results were seen). We registered this as the primary hypothesis of Step 10 and retrained E and the SAR-only U-Net on seeds 11–20, which no arm had used:

Table 6. Step 10, seeds 11–20, 10 test cities (primary: single test; secondary: Holm over 2).

contrast	mean Δ	95% CI	d_z	Δ > 0	Holm p	label
E@no_opt – SAR-only	+0.072	[+0.038, +0.107]	+1.48	10/10	0.0011	E without optical higher
E: image present – absent	−0.116	[−0.145, −0.086]	−2.76	0/10	2.2e-05	lower with the image
E@no_opt – SAR-only, threshold 0.5	+0.109	[+0.070, +0.148]	+2.00	10/10	1.4e-04	E without optical higher

The gain replicates over training seeds (E 0.495 ± 0.012, SAR-only 0.423 ± 0.045) and holds at a fixed threshold, so it is not only a threshold effect. B, trained with the same mixture but late fusion, shows a smaller gain without optical input (+0.022 at its `no_opt` threshold, 95% CI [−0.003, +0.048], exploratory). The new seeds replicate training, not cities: in the exploratory city analysis the gain is concentrated in Las Vegas (+0.157 in that city); without it the pooled gain is +0.028, it stays positive with any one city left out (+0.028 to +0.082), and only 5 of 10 cities share its sign. The mechanism, for example regularisation by the optical inputs during training, was not tested.

4.4 RQ2: acquisition timing of the historical optical image

Under OSCD's label definition (change between two fixed dates), varying the image's timing varies two things at once: its age and how much of the labelled change it may already show, because every image at 1–12 months post-dates OSCD date 1. B, setting `no_o2`, test F1 at the frozen `no_o2` threshold, 9 primary cities (Step 8):

arm	OSCD date 1	Earth Engine date 1	1 month	3 months	6 months	12 months	18 months
B dropout	0.420 ± 0.039	0.420 ± 0.039	0.389 ± 0.044	0.422 ± 0.038	0.409 ± 0.035	0.417 ± 0.037	0.412 ± 0.033
async	0.405 ± 0.067	0.407 ± 0.066	0.349 ± 0.059	0.416 ± 0.061	0.370 ± 0.072	0.422 ± 0.057	0.383 ± 0.066

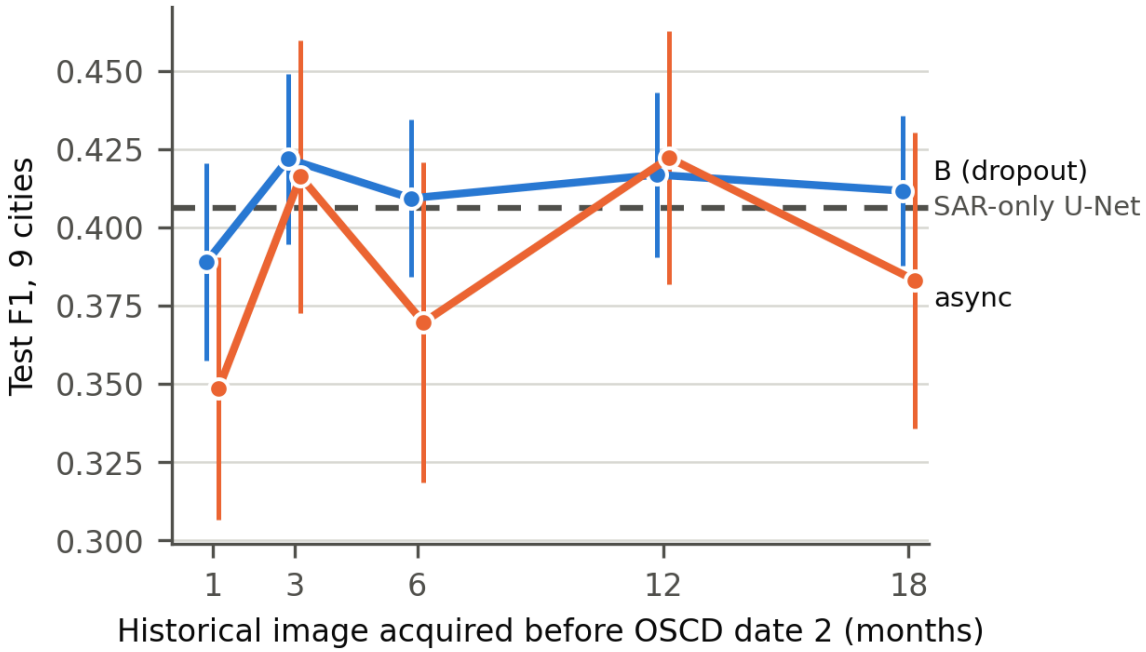


Figure 2. Test F1 by acquisition timing of the historical image (mean and 95% CI over 10 seeds, 9 cities); dashed: SAR-only U-Net on the same cities.

Primary test: the per-seed least-squares slope of B's F1 against the gap has mean +0.0083 F1 per year (95% CI [−0.0021, +0.0187]), t-test p 0.105: **no detectable linear trend**. Against SAR alone (0.406 ± 0.048 on these cities), B shows no detectable difference at any timing. **Exploratory:** the timings do differ (Friedman p 0.00062 for B), with the 1-month image lowest for both models; B's 1-month deficit relative to OSCD's date-1 image comes mostly from Saclay and is not established when cities are resampled (95% CI [−0.116, +0.003]). Each timing has one scene per city, so scene properties are not replicated by seeds. In Stage 4B's descriptive comparison, no timing scored detectably above B without any optical input (largest difference +0.004), and the 1-month image scored below it (−0.029, uncorrected p 0.041).

Answer to RQ2. No detectable linear trend of F1 with acquisition time over 1–18 months, and no timing at which the image beats SAR alone. The low 1-month score is consistent with the image already showing part of the change, but this was not measured, so the design does not isolate image age. Separating age from the labelled change requires labels dated at the historical image. The decline reported by Song et al. [2] concerns a different role of the image and is not tested here.

4.5 RQ3: VV and VH polarisation

Full input, test F1 at the frozen threshold, 9 test cities with VH at both dates (Step 7):

arm	F1	precision	recall	IoU	false-alarm rate
SAR-only U-Net, VV	0.452 ± 0.011	0.367 ± 0.017	0.590 ± 0.029	0.292 ± 0.009	0.049 ± 0.005
SAR-only U-Net, VH	0.383 ± 0.029	0.346 ± 0.048	0.442 ± 0.069	0.237 ± 0.022	0.042 ± 0.016
SAR-only U-Net, VV+VH	0.408 ± 0.031	0.405 ± 0.040	0.424 ± 0.081	0.257 ± 0.024	0.031 ± 0.010
DS-UNet, VV	0.501 ± 0.024	0.523 ± 0.067	0.500 ± 0.089	0.335 ± 0.021	0.024 ± 0.013
DS-UNet, VH	0.484 ± 0.043	0.511 ± 0.030	0.462 ± 0.061	0.320 ± 0.036	0.021 ± 0.002
DS-UNet, VV+VH	0.511 ± 0.019	0.571 ± 0.029	0.465 ± 0.036	0.344 ± 0.017	0.017 ± 0.003

contrast	mean Δ	95% CI	d_z	Δ > 0	Holm p	label
SAR-only: VV+VH – VV	−0.044	[−0.065, −0.023]	−1.49	1/10	0.0011	VV+VH lower
SAR-only: VH – VV	−0.070	[−0.091, −0.048]	−2.32	0/10	8.8e-05	VH lower
DS-UNet: VV+VH – VV	+0.010	[−0.016, +0.036]	+0.27	7/10	0.4125	no detectable difference
DS-UNet: VH – VV	−0.017	[−0.039, +0.004]	−0.59	3/10	0.1925	no detectable difference

The first two rows are the primary family, the last two a secondary family. By land-cover class, the SAR-only VV+VH model has lower false-alarm rates than VV in four classes but lower recall in every class (0.424 against 0.590 overall): it flags less change everywhere, a shift of operating point, not evidence that VH separates urban development from pseudo-change. Including Las Vegas, whose VH band at date 1 is empty, the VV+VH SAR-only model drops to F1 0.224 ± 0.032.

Answer to RQ3. For the SAR-only U-Net, adding VH to VV lowered F1 and VH alone lowered it more; with optical input, polarisation made no detectable difference. These are results for 3-month mean backscatter whose date-1 means average fewer scenes, and OSCD has no labels for the cause of a false alarm.

4.6 RQ5: conditions in which historical optical information helps

False-alarm rate by WorldCover class, B with OSCD's date-1 image minus the SAR-only U-Net, 9 primary cities (Step 8; Holm over the 7 classes with ≥ 1% of the pixels):

class	pixel share	mean Δ	95% CI	Δ > 0	Holm p	label
tree cover	15.6%	−0.001	[−0.003, −0.000]	3/10	0.0973	no detectable difference
shrubland	1.3%	−0.006	[−0.010, −0.003]	1/10	0.0127	B lower
grassland	7.7%	+0.002	[−0.002, +0.006]	5/10	0.9703	no detectable difference
cropland	8.0%	−0.015	[−0.034, +0.004]	2/10	0.5354	no detectable difference
built-up	40.6%	−0.004	[−0.015, +0.008]	4/10	0.9804	no detectable difference
bare	24.6%	−0.025	[−0.077, +0.027]	4/10	0.9703	no detectable difference
water	1.7%	+0.001	[−0.008, +0.011]	4/10	0.9804	no detectable difference

The one detectable difference, in shrubland (1.3% of pixels), is between two networks. Within B (Stage 4B), the historical image does not detectably change shrubland false alarms (Holm p 0.28), so that difference cannot be attributed to the image. Within B the image lowers the tree-cover false-alarm rate by 0.002 (Holm p 0.005) and raises recall in tree cover and built-up areas (+0.020 each); these effects are small.

Answer to RQ5. Within the tested conditions (1–18 month timings, seven land-cover classes, OSCD urban change), we found no condition in which the historical image gives a material benefit.

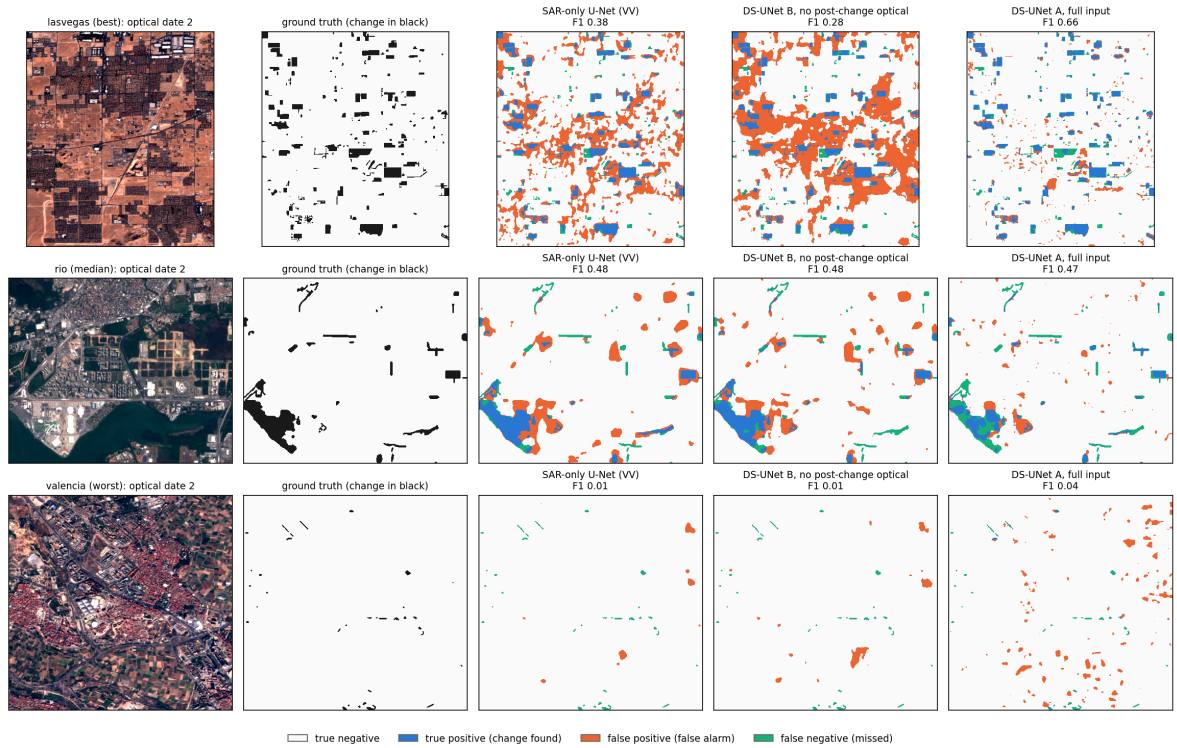


Figure 3. Error maps for the best (Las Vegas), median (Rio) and worst (Valencia) OSCD test city by DS-UNet A's mean per-city F1; seed 1, frozen thresholds. Blue: true positive; orange: false positive; green: false negative.

4.7 Test cities and training seeds (exploratory)

Per-city counts at the same frozen thresholds; the first interval resamples seeds, the second resamples test cities and then seeds (two-stage percentile bootstrap, 10000 replicates). Dubai, Las Vegas and Chongqing hold 77% of the change pixels.

contrast	cities	mean Δ	95% CI, seeds	SE, seeds	SE, cities	cities with same sign	95% CI, cities and seeds	Δ, one city left out
A (full input) – SAR-only	10	+0.138	[+0.094, +0.181]	0.019	0.087	6/10	[−0.022, +0.246]	+0.049 to +0.156
B@no_02, own threshold – SAR-only	10	+0.013	[−0.015, +0.040]	0.012	0.004	6/10	[−0.019, +0.043]	+0.009 to +0.014
async@no_02 – SAR-only	10	−0.016	[−0.059, +0.026]	0.019	0.039	7/10	[−0.095, +0.047]	−0.055 to −0.001
B@no_02, clean threshold – SAR-only	10	+0.060	[+0.026, +0.094]	0.015	0.037	6/10	[+0.007, +0.117]	+0.024 to +0.073
B: 1-month – OSCD date-1 image	9	−0.031	[−0.056, −0.005]	0.011	0.026	5/9	[−0.116, +0.003]	−0.043 to −0.008

For four of the five contrasts, resampling cities adds more uncertainty than resampling seeds. DS-UNet's lead over the SAR-only U-Net is concentrated in Las Vegas (without it +0.049) and its city-and-seed interval includes zero; this shows that ten cities do not establish the lead beyond them, not that it is absent. Las

Vegas also carries most of the Step 10 gain (§4.3): with 10 cities of very unequal size, a single large desert city can drive pooled differences.

5. Missing, synthetic-cloud and real-cloud post-change images

RQ1 assumes that the post-change optical image is unavailable; in practice it is often present but cloudy. All corruptions below act on the post-change Sentinel-2 image only, and models use their frozen full-input threshold (a deployed model does not know the cloud level). Zero is the fill value used here, not an intrinsic missing-data token.

5.1 Whole-modality dropout handles a missing image at no detectable full-input cost (Phase 2)

Test F1 at the validation-selected threshold (mean ± SD over 10 seeds; primary family B − A, Holm over 4):

condition	A full-only	B dropout	Δ (B − A)	95% CI	B > A	Holm p
full	0.556 ± 0.029	0.553 ± 0.030	−0.003	[−0.037, +0.031]	5/10	0.8449
post optical missing (`no_o2`)	0.190 ± 0.068	0.431 ± 0.034	+0.241	[+0.176, +0.305]	10/10	5.8e-05
copy-forward (`o2_copy`)	0.314 ± 0.034	0.438 ± 0.038	+0.124	[+0.087, +0.160]	10/10	9.5e-05
no optical (`no_opt`)	0.397 ± 0.106	0.440 ± 0.027	+0.043	[−0.036, +0.123]	5/10	0.4964

B is better with the post-change image missing or replaced by the pre-change image. On full input there is no detectable cost, although losses up to 0.037 F1 are compatible with the data; at a fixed threshold of 0.5, B is lower (−0.086), so the dropout model needs its threshold chosen on validation data.

5.2 Partial synthetic cloud is a different failure, and dropout does not fix it (Phase 3)

Clouds are real catalogue masks [14] scaled to 10 / 30 / 60% of each city, filled with an opaque, uniform cloud spectrum. Test F1 at the frozen clean threshold (B − A, Holm over 4):

condition	A full-only	B dropout	Δ (B − A)	Holm p	B > A
clean	0.556 ± 0.029	0.553 ± 0.030			
c10	0.285 ± 0.015	0.329 ± 0.008	+0.044	9.6e-05	10/10
c30	0.165 ± 0.007	0.191 ± 0.012	+0.026	0.0010	10/10
c60	0.123 ± 0.002	0.134 ± 0.011	+0.011	0.0144	10/10
missing	0.142 ± 0.050	0.478 ± 0.030	+0.336	9.5e-08	10/10

For B, a partly cloudy post-change image is far worse than none (0.329 at 10% cover against 0.478 missing): almost every pixel under a cloud is predicted as change (false-alarm rate inside the mask 0.96–1.00), and false alarms spill into nearby clear pixels. Dropout taught B to ignore zero-filled optical input, not cloud-valued input.

5.3 Zero-filling under the cloud recovers most of the loss, mainly for the dropout model (Step 4A)

arm	level	F1 cloud	F1 zero-fill	Δ	recovery of the cloud loss
A	10%	0.285	0.393	+0.108	0.403 ± 0.189
A	30%	0.165	0.261	+0.097	0.249 ± 0.168
A	60%	0.123	0.192	+0.069	0.161 ± 0.132
B	10%	0.329	0.539	+0.210	0.939 ± 0.016
B	30%	0.191	0.516	+0.325	0.899 ± 0.033
B	60%	0.134	0.499	+0.365	0.872 ± 0.041

With the true mask, zero-fill returns B's in-mask false-alarm rate to its clean value; A reads the zero-filled region as change too. The remedy works well only together with dropout training.

5.4 Mask errors, spatial dropout and a real detector on composites (Steps 4A-bis–5)

Four further steps, detailed in Appendix E, qualify §5.3. With synthetic mask errors at 30% cover (Table E1), over-masking is almost free for B (masking 21% of the clear pixels costs 0.010 F1) while under-masking is expensive (missing a quarter of the cloud costs 0.168). Training with cloud-shaped zero masks (arm C, Table E2) gave no detectable gain over B in any of 7 primary comparisons. Dilating an under-masking mask repairs missed cloud edges but not missed whole clouds (Table E3). With real catalogue cloud pixels pasted into the post-change images and masked by s2cloudless at operating points chosen on validation composites (Table E4), the recall-biased point rb95 beat the most accurate point at 30% and 60% cover, but it does so by masking 0.596–0.673 of the clear pixels, so its gain is consistent with B falling back on SAR and the pre-change image rather than with good cloud localisation.

5.5 Real cloudy scenes (Step 11)

Step 11 evaluates the frozen A, B and SAR-only models on the real scenes of §3.1, masked by s2cloudless at Step 5's frozen operating points. Before any change model saw a scene, the detector outputs showed that rb99 masks essentially the whole image and that rb95 masks 0.10–0.98 of the clear control scenes, so the registered analysis compares masking with discarding the image.

Table 7. Step 11, 7 partly cloudy cities (city-box cloud 0.08–0.68; scenes 3–15 days from OSCD date 2), F1 at the frozen full-input thresholds (primary family Holm over 3; secondary Holm over 4).

contrast	mean Δ	95% CI	d_z	Δ > 0	Holm p	label
B: rb95 mask – no mask	+0.208	[+0.179, +0.236]	+5.22	10/10	1.5e-07	higher with the mask
B with rb95 mask – SAR-only	+0.020	[+0.003, +0.036]	+0.86	9/10	0.0242	B higher
B – A, no mask	+0.087	[+0.061, +0.113]	+2.36	10/10	7.6e-05	B higher
B: bal mask – no mask	+0.209	[+0.182, +0.236]	+5.54	10/10	8.7e-08	higher with the mask
B: rb95 mask – image discarded	−0.007	[−0.012, −0.002]	−0.94	2/10	0.0200	lower with the mask
A, no mask – SAR-only	−0.275	[−0.291, −0.259]	−12.55	0/10	8.2e-11	A lower
B: Earth Engine control – OSCD image	−0.002	[−0.003, −0.001]	−1.03	1/10	0.0200	control lower

Table 8. Step 11, test F1 on the 7 partly cloudy cities (mean ± SD over 10 seeds; descriptive).

post-change input	A full-only	B dropout
OSCD image (clear)	0.488 ± 0.030	0.527 ± 0.021
Earth Engine control (clear, same day)	0.481 ± 0.028	0.525 ± 0.021
real cloudy scene, no mask	0.162 ± 0.015	0.248 ± 0.041
real cloudy scene, bal mask	0.164 ± 0.053	0.457 ± 0.033
real cloudy scene, rb95 mask	0.133 ± 0.045	0.456 ± 0.016
image discarded	0.133 ± 0.056	0.463 ± 0.011

The SAR-only U-Net, which ignores optical input, scores 0.436 ± 0.011 on these cities. Real cloud reproduces the synthetic result: passed through unmasked, it drops A far below SAR alone and B to 0.248. With cloud pixels zero-filled, B recovers most of the loss and scores slightly above the SAR-only fallback, a margin that stays positive with any one city left out (+0.010 to +0.022). Masking did not beat discarding the image (−0.007), and B with the image discarded also scored above SAR-only (+0.026, exploratory). On the 3 overcast cities and on all 10, the three primary contrasts have the same sign and are detectable (for example, B with the rb95 mask − SAR-only +0.091 and +0.056). The processing control differs from OSCD's own image by −0.002. Each city has one real scene, cloud cover is not controlled, and the Las Vegas "overcast" scene is probably bright desert flagged by the detector rather than opaque cloud.

6. Discussion

What the historical image does. Without the post-change optical image, the post-change information comes from SAR, and we found no added change evidence in the pre-change optical image. The early-fusion experiment rules out the most obvious explanation of the DS-UNet null, that late fusion cannot use the image: given the chance to condition the SAR features, the image hurt. The rise in false alarms suggests that, in a bitemporal input, an optical image at date 1 next to zeros at date 2 looks like change, and that dropout training did not remove this reading. A model that uses historical imagery as Song et al. [2] do, as a separate semantic prior rather than as one date of a pair, may behave differently; our results bear on the common practice of filling the pre-change slot of a bitemporal model.

What dropout training does. Modality dropout made two models usable without the post-change image. With no optical input, the early-fusion model scored detectably above a U-Net trained on SAR alone, replicated on new seeds; the late-fusion B scored above it too, but not detectably (+0.022, 95% CI [−0.003, +0.048]). Training with optical inputs that are sometimes removed may act as a regulariser or as a form of cross-modal supervision of the SAR pathway; we did not test the mechanism, and the gain is concentrated in Las Vegas.

Missing and cloudy are different problems. A model trained only with full input reads cloud, or zero-filled cloud, as change; on real scenes it fell 0.275 below SAR alone. A dropout-trained model turns the cloud problem back into the missing-data problem once cloud pixels are zero-filled, and the outcome then depends on the direction of the mask error: over-masking is cheap, under-masking is expensive. On real scenes the detector's recall-biased masks removed most of the image, and discarding the image did at least as well. For this configuration the practical rule is: train with modality dropout, and when cloud is detected over the scene, drop the post-change optical image rather than pass it through.

Polarisation and operating point. For SAR alone, VV gave the highest F1 on OSCD and VH made the model more conservative; a missing VH acquisition nearly halved the VV+VH model's pooled F1. Dropout-trained and U-Net baseline models shift their output probabilities and need thresholds chosen on validation data, and the threshold rule can change a comparison (B's margin over SAR-only is +0.013 at its `no_o2`

threshold and +0.060 at its full-input threshold). Deployments should select thresholds on held-out data from the target region.

7. Limitations

- **One benchmark, few cities.** Results use 9 or 10 OSCD test cities and 4 validation cities. Seeds measure training variability on fixed cities; several effects (A's lead over SAR-only, the Step 10 gain) are concentrated in Las Vegas.
- **Power.** Effects below about 1.00–1.22 SD of the paired differences can be missed; nulls are not equivalence.
- **No pseudo-change labels.** RQ3 and RQ5 are answered through false-alarm rates by land-cover class, a stratification rather than a label of the cause.
- **Temporal-gap confound.** Image age is confounded with how much labelled change the image shows; one scene per city and timing.
- **Architectures and recipe.** Two fusion designs (late-fusion DS-UNet, early-fusion U-Net), one recipe tuned on DS-UNet, untuned baselines with thresholds near the top of the grid; reconstruction- or attention-based models [2, 12] may differ.
- **SAR input.** 3-month mean backscatter; date-1 means average fewer scenes.
- **Clouds.** Synthetic and composite clouds are opaque and shadowless; the real-cloud test has one scene per city, 3–15 days from the label date, uncontrolled cover and one detector.
- **Internal pre-registration.** Pre-registrations are repository files; Stage 4B followed partial unblinding, the Step 10 hypothesis came from Step 9's results, and the same AI assistant wrote analysis code, audits and simulated reviews.

8. Conclusion

Under one multi-seed protocol on OSCD: (RQ1) a historical optical image did not improve SAR change detection without the post-change optical image, making no detectable difference in a late-fusion DS-UNet and lowering F1 in an early-fusion U-Net; (RQ2) its acquisition time over 1–18 months showed no detectable linear trend, with a confounded 1-month low; (RQ3) adding VH lowered F1 for the SAR-only U-Net; (RQ4) without the post-change image neither B nor async was detectably better than SAR alone in the registered comparison at per-condition thresholds (B's margin at its full-input threshold comes from B itself, not the historical image), while with full input DS-UNet scored highest across seeds, a lead concentrated in one city; (RQ5) no land-cover or timing condition showed a material benefit. Modality dropout gave a SAR-only detector better than SAR-only training (early fusion, concentrated in one city). On real clouds, a dropout-trained model with the cloud masked scored slightly above SAR alone (registered), discarding the cloudy image did at least as well, and a model trained only on full input fell far below SAR alone. Whether historical imagery helps as a separate semantic prior, with labels dated to separate age from change, or on other benchmarks, remains open.

Declarations

Data availability. OSCD [3] and the Cloud Mask Catalogue [14] are public; the Sentinel-1 VV rasters are from the public DS-UNet repository [1]; Sentinel-1, Sentinel-2 (historical and real-cloud scenes) and ESA WorldCover [19] were exported from Google Earth Engine [18] with the scripts in the project repository. Code, pre-registrations, run files, analysis tables, audits and reports are public at https://github.com/Manishym956/Multimodal-Satellite-Change-Detection-for-Urban-Development-and-Environmental-Monitoring (branch `main`; code under the GNU GPL, version 2 or later). Checkpoints,

downloaded scenes and raw count arrays are not in the repository because of their size; they are identified by SHA-256 hashes in the pre-registrations and `experiments/robust/ARTIFACT_SHA256_*.txt` and are available from the author on request.

Ethics. The study uses public satellite imagery and labels only; no human participants or personal data are involved.

Author contributions (CRediT). [author: to be agreed with the mentor. Draft: Manish Y. M.: conceptualization, methodology, software, investigation, formal analysis, data curation, visualization, writing (original draft and editing); Soumen Mukherjee: supervision, writing (review).]

Conflict of interest. [author: confirm.] The author declares no conflict of interest.

Funding. [author: confirm.] No external funding.

Use of AI tools. An AI assistant (Claude, Anthropic) helped write analysis and evaluation code, audit scripts, report builders and figures, and drafted parts of the text. All results come from the code and outputs in the repository; the author checked and is responsible for the content. The same assistant ran the pre-submission integrity checks and simulated peer review, so those checks are not independent of the drafting.

References

[1] Hafner, S., Nascetti, A., Azizpour, H., & Ban, Y. (2022). Sentinel-1 and Sentinel-2 data fusion for urban change detection using a dual stream U-Net. *IEEE Geoscience and Remote Sensing Letters, 19*, 1–5. https://doi.org/10.1109/LGRS.2021.3119856

[2] Song, W., Lv, Q., Ding, Z., Hong, S., & Qi, Z. (2026). A dual-modal framework integrating SAR-based change screening and optical-scene-informed identification for high-frequency monitoring of construction-ready bare land. *Remote Sensing, 18*(8), 1103. https://doi.org/10.3390/rs18081103

[3] Caye Daudt, R., Le Saux, B., Boulch, A., & Gousseau, Y. (2018). Urban change detection for multispectral earth observation using convolutional neural networks. In *Proc. IEEE IGARSS*, 2115–2118. https://doi.org/10.1109/IGARSS.2018.8518015

[4] Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics, 6*(2), 65–70. https://www.jstor.org/stable/4615733

[5] Caye Daudt, R., Le Saux, B., & Boulch, A. (2018). Fully convolutional siamese networks for change detection. In *Proc. IEEE ICIP*, 4063–4067. https://doi.org/10.1109/ICIP.2018.8451652

[6] Ronneberger, O., Fischer, P., & Brox, T. (2015). U-Net: Convolutional networks for biomedical image segmentation. In *MICCAI*, LNCS 9351, 234–241. https://doi.org/10.1007/978-3-319-24574-4_28

[7] Shafique, A., Cao, G., Khan, Z., Asad, M., & Aslam, M. (2022). Deep learning-based change detection in remote sensing images: A review. *Remote Sensing, 14*(4), 871. https://doi.org/10.3390/rs14040871

[8] Ebel, P., Saha, S., & Zhu, X. X. (2021). Fusing multi-modal data for supervised change detection. *International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences, XLIII-B3-2021*, 243–249. https://doi.org/10.5194/isprs-archives-XLIII-B3-2021-243-2021

[9] Saha, S., Ebel, P., & Zhu, X. X. (2022). Self-supervised multisensor change detection. *IEEE Transactions on Geoscience and Remote Sensing, 60*, 4405710. https://doi.org/10.1109/TGRS.2021.3109957

[10] Jiang, J., Xing, Y., Wei, W., Yan, E., Xiang, J., & Mo, D. (2022). DSNUNet: An improved forest change detection network by combining Sentinel-1 and Sentinel-2 images. *Remote Sensing, 14*(19), 5046. https://doi.org/10.3390/rs14195046

[11] Neverova, N., Wolf, C., Taylor, G. W., & Nebout, F. (2016). ModDrop: Adaptive multi-modal gesture recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence, 38*(8), 1692–1706. https://doi.org/10.1109/TPAMI.2015.2461544

[12] Hafner, S., & Ban, Y. (2023). Multi-modal deep learning for multi-temporal urban mapping with a partly missing optical modality. In *Proc. IEEE IGARSS*, 6843–6846. https://doi.org/10.1109/IGARSS52108.2023.10281626

[13] Ebel, P., Meraner, A., Schmitt, M., & Zhu, X. X. (2021). Multisensor data fusion for cloud removal in global and all-season Sentinel-2 imagery. *IEEE Transactions on Geoscience and Remote Sensing, 59*(7), 5866–5878. https://doi.org/10.1109/TGRS.2020.3024744

[14] Francis, A., Mrziglod, J., Sidiropoulos, P., & Muller, J.-P. (2020). Sentinel-2 Cloud Mask Catalogue (Version 1). Zenodo. https://doi.org/10.5281/zenodo.4172871

[15] Sinergise EO research team (2024). s2cloudless: Sentinel Hub's cloud detector for Sentinel-2 imagery (Version 1.7.3) [Software]. https://github.com/sentinel-hub/sentinel2-cloud-detector

[16] Skakun, S., Wevers, J., Brockmann, C., Doxani, G., Aleksandrov, M., Batič, M., Frantz, D., Gascon, F., Gómez-Chova, L., Hagolle, O., López-Puigdollers, D., Louis, J., Lubej, M., Mateo-García, G., Osman, J., Peressutti, D., Pflug, B., Puc, J., Richter, R., Roger, J.-C., Scaramuzza, P., Vermote, E., Vesel, N., Zupanc, A., & Žust, L. (2022). Cloud Mask Intercomparison eXercise (CMIX): An evaluation of cloud masking algorithms for Landsat 8 and Sentinel-2. *Remote Sensing of Environment, 274*, 112990. https://doi.org/10.1016/j.rse.2022.112990

[17] Bouthillier, X., Delaunay, P., Bronzi, M., Trofimov, A., Nichyporuk, B., Szeto, J., et al. (2021). Accounting for variance in machine learning benchmarks. In *Proceedings of Machine Learning and Systems (MLSys), 3*.

[18] Gorelick, N., Hancher, M., Dixon, M., Ilyushchenko, S., Thau, D., & Moore, R. (2017). Google Earth Engine: Planetary-scale geospatial analysis for everyone. *Remote Sensing of Environment, 202*, 18–27. https://doi.org/10.1016/j.rse.2017.06.031

[19] Zanaga, D., et al. (2021). ESA WorldCover 10 m 2020 v100. Zenodo. https://doi.org/10.5281/zenodo.5571936

[20] Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., et al. (2021). An image is worth 16×16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations (ICLR)*.

[21] Bandara, W. G. C., & Patel, V. M. (2022). A transformer-based Siamese network for change detection. In *Proc. IEEE IGARSS*, 207–210. https://doi.org/10.1109/IGARSS46834.2022.9883686

[22] Chen, H., Qi, Z., & Shi, Z. (2022). Remote sensing image change detection with transformers. *IEEE Transactions on Geoscience and Remote Sensing, 60*, 1–14. https://doi.org/10.1109/TGRS.2021.3095166

[23] Van Etten, A., Hogan, D., Martinez Manso, J., Shermeyer, J., Weir, N., & Lewis, R. (2021). The multi-temporal urban development SpaceNet dataset. In *Proc. IEEE/CVF CVPR*, 6398–6407.

https://doi.org/10.1109/CVPR46437.2021.00633

[24] Schmitt, M., Hughes, L. H., Qiu, C., & Zhu, X. X. (2019). SEN12MS – A curated dataset of georeferenced multi-spectral Sentinel-1/2 imagery for deep learning and data fusion. *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences, IV-2/W7*, 153–160. https://doi.org/10.5194/isprs-annals-IV-2-W7-153-2019

[25] Liu, J., Gong, M., Qin, K., & Zhang, P. (2018). A deep convolutional coupling network for change detection based on heterogeneous optical and radar images. *IEEE Transactions on Neural Networks and Learning Systems, 29*(3), 545–559. https://doi.org/10.1109/TNNLS.2016.2636227

[26] Luppino, L. T., Kampffmeyer, M., Bianchi, F. M., Moser, G., Serpico, S. B., Jenssen, R., & Anfinsen, S. N. (2022). Deep image translation with an affinity-based change prior for unsupervised multimodal change detection. *IEEE Transactions on Geoscience and Remote Sensing, 60*, 1–22. https://doi.org/10.1109/TGRS.2021.3056196

[27] Chen, H., Song, J., Dietrich, O., Broni-Bediako, C., Xuan, W., Wang, J., et al. (2025). BRIGHT: A globally distributed multimodal building damage assessment dataset with very-high-resolution for all-weather disaster response. *Earth System Science Data, 17*(11), 6217–6253. https://doi.org/10.5194/essd-17-6217-2025

[28] Ebel, P., Xu, Y., Schmitt, M., & Zhu, X. X. (2022). SEN12MS-CR-TS: A remote-sensing data set for multimodal multitemporal cloud removal. *IEEE Transactions on Geoscience and Remote Sensing, 60*, 1–14. https://doi.org/10.1109/TGRS.2022.3146246

Appendix A: Preliminary study under a simpler protocol

Before the shared protocol of §3, a preliminary study trained re-implemented models with a simpler recipe (BCE loss, Adam 1e-4, 50 epochs, no augmentation or importance sampling, threshold 0.5) and 3–5 seeds per model on the same split. Its dual-stream model is a re-implementation, not the official DS-UNet; its missing-optical models are dual-stream networks trained without the post-change optical image (dropping those channels, “clean”, or zero-filling them, “naive”); and it includes two lightweight Transformers [20] (1.16–1.36M parameters), motivated by Transformer change detectors [21, 22]. Its results are superseded by §4 and are reported for completeness.

Table A1: OSCD test results of the preliminary study (10 test cities, τ = 0.5; mean ± sample SD over n seeds)

Model	Loss	n	F1	Precision	Recall	IoU	FAR
SAR-only U-Net	pw=1	5	0.241 ± 0.101	0.221 ± 0.152	0.342 ± 0.074	0.140 ± 0.065	0.112 ± 0.106
Optical-only U-Net	pw=1	3	0.204 ± 0.055	0.393 ± 0.176	0.167 ± 0.101	0.114 ± 0.035	0.021 ± 0.022
Early Fusion U-Net	pw=1	3	0.376 ± 0.022	0.343 ± 0.081	0.439 ± 0.070	0.232 ± 0.017	0.050 ± 0.021
Sync Dual-Stream CNN	pw=1	5	0.459 ± 0.061	0.420 ± 0.079	0.518 ± 0.074	0.299 ± 0.051	0.041 ± 0.017
Naive Missing-Optical CNN	pw=1	3	0.297 ± 0.089	0.307 ± 0.132	0.324 ± 0.141	0.177 ± 0.061	0.047 ± 0.035
Clean Missing-Optical CNN	pw=1	5	0.309 ± 0.095	0.428 ± 0.154	0.254 ± 0.075	0.186 ± 0.066	0.023 ± 0.019
Transformer Self-Attn (original)	pw=41	3	0.199 ± 0.048	0.117 ± 0.029	0.679 ± 0.178	0.111 ± 0.030	0.283 ± 0.070
Transformer Cross-Attn (original)	pw=41	3	0.275 ± 0.124	0.190 ± 0.103	0.632 ± 0.116	0.163 ± 0.080	0.240 ± 0.255
Transformer Cross-Attn (pw=1)	pw=1	3	0.088 ± 0.012	0.053 ± 0.001	0.408 ± 0.348	0.046 ± 0.007	0.399 ± 0.345
Transformer Self-Attn (tuned, wd=1e-4)	pw=41	3	0.212 ± 0.024	0.125 ± 0.017	0.695 ± 0.018	0.118 ± 0.015	0.269 ± 0.047
Transformer Cross-Attn (tuned, wd=1e-4)	pw=41	3	0.231 ± 0.065	0.142 ± 0.048	0.723 ± 0.168	0.131 ± 0.041	0.285 ± 0.201

pw = BCE pos_weight. "Original" Transformers use weight decay 0; "tuned" use weight decay 1e-4 selected on validation F1.

Table A2: Pairwise F1 comparisons of the preliminary study (two-sided Welch t-test, Holm over 6)

Comparison (F1)	Δ mean	n	p (Welch)	p (Holm)
Sync dual-stream – Clean missing-optical	+0.150	5 vs 5	0.022	0.109
Clean missing-optical – SAR-only	+0.069	5 vs 5	0.302	0.604
Sync dual-stream – SAR-only	+0.218	5 vs 5	0.005	0.030
Sync dual-stream – Early fusion	+0.083	5 vs 3	0.037	0.148
Clean – Naive missing-optical	+0.012	5 vs 3	0.865	0.865
Transformer Cross-Attn pw=41 – pw=1	+0.186	3 vs 3	0.119	0.358

Three findings carry over as context. First, **seed variance changes the ranking**: seed-to-seed SDs of 0.06–0.12 F1 were common, and the first seed alone ranked the clean missing-optical model (0.412) above the synchronous model (0.387), the opposite of the five-seed mean; this motivated the 10-seed design of §3. Second, under this weaker recipe every baseline scored far lower than under the shared protocol (SAR-only 0.241 against 0.418 in Step 6), so its RQ1 margin (0.309 against 0.241, not significant) is not evidence. Third, the cross-attention Transformer reached unimodal-CNN-level F1 only with strong class reweighting (pw=41)

and collapsed to predicting no change with pw=1; with one small design per attention type, this is a finding about these models under this recipe, not about Transformers in general.

Table A1 uses τ = 0.5. Table A3 also reports each Transformer run at its own threshold. That threshold was the argmax of pooled validation F1 on the 4 validation cities, over a grid from 0.05 to 0.8 refined near the peak, and no test score selected a threshold. For the original checkpoints, only seed 42 was re-evaluated this way.

Table A3: Transformers at validation-selected thresholds (OSCD, 10 test cities; mean ± sample SD over n seeds)

Model	n	Selected τ (seeds 42 / 43 / 44)	F1 at τ = 0.5	F1	Precision	Recall	IoU	FAR
Transformer Self-Attn (original, seed 42)	1	0.6125	0.162	0.156	0.096	0.406	0.085	0.207
Transformer Self-Attn (tuned, wd=1e-4)	3	0.4375 / 0.4875 / 0.575	0.212 ± 0.024	0.213 ± 0.011	0.127 ± 0.006	0.669 ± 0.093	0.119 ± 0.007	0.251 ± 0.032
Transformer Cross-Attn (original, seed 42)	1	0.5625	0.356	0.385	0.321	0.481	0.238	0.055
Transformer Cross-Attn (tuned, wd=1e-4)	3	0.525 / 0.575 / 0.725	0.231 ± 0.065	0.280 ± 0.040	0.187 ± 0.034	0.563 ± 0.023	0.163 ± 0.028	0.136 ± 0.022

F1, precision, recall, IoU and FAR are at the selected τ. Original-checkpoint rows are seed 42 only, so they differ from the 3-seed means in Table A1.

Selecting the threshold on validation data raised the tuned cross-attention model's mean test F1 from 0.231 to 0.280, mostly by cutting false alarms (FAR 0.285 to 0.136). It left self-attention essentially unchanged (0.212 to 0.213). The selected thresholds varied widely between seeds (0.525–0.725 for cross-attention), and with n = 3 none of these differences is tested for significance. The best Transformer result here (0.385, original cross-attention, one seed) remains below every model in §4.1 (lowest: SAR-only U-Net, 0.418), though under a different training recipe.

Appendix B: SEN12-MT note

The preliminary study also trained models on one 21-month SAR–optical interval of a pre-packaged Sentinel-1/2 archive over SpaceNet 7 [23] sites (distinct from SEN12MS [24]), with proxy labels from building footprints and 4 test sites; all models scored F1 0.08–0.11 with one seed. The archive's provenance and metadata could not be verified, so these results are not used. RQ2 is answered by Step 8 on OSCD (§4.3).

Appendix C: Deviations and gaps

1. **Phase 2 step budget.** A 1000-step budget was first proposed after the test F1 of a 3-seed step sweep had been viewed. That choice was discarded and the budget re-chosen on validation score (2500

steps). Seeds 1–3 of A share the configuration of a sweep run whose test F1 had been seen; they were retrained from scratch, reproduced identically, and no Phase 2 decision used them.

2. **Phase 3 masks.** The first mask build used an absolute cloud-fraction tolerance, which could nearly double a small cloud; the masks were rebuilt with a relative tolerance before any model saw a mask. A missing `model.eval()` in the Phase 3 evaluation script was fixed after a validation-only smoke test, before any test evaluation.
3. **Steps 4A and 4A-bis.** Both pre-registrations cite a validation-only smoke test for which no log exists. In Step 4A-bis, the evaluation printed per-seed test F1 while running; the analysis crashed on its final console print and was re-run unchanged; two source files hashed in every run file were missing from the pre-registration's hash table (unchanged since Phase 3). Inside/outside-mask metrics and IoU were added as a labelled exploratory addendum.
4. **Phase 3 report wording.** `PHASE3_REPORT.md` calls the all-change F1 the "change prevalence"; the prevalence is 0.052. The correction is in `experiments/robust/PHASE3_REPORT_ERRATA.md`.
5. **Step 4B.** Test F1 was printed at the end of each training run, as pre-specified. The pre-registration says "33" evaluation conditions but lists 34; all 34 were evaluated.
6. **Step 5S.** The validation smoke test's script was not committed, so the smoke test cannot be re-run from the repository.
7. **Step 5.** The scene pools were redesigned before any composite was built (arm C reported as flagged); the composite build was interrupted twice and resumed (89 files verified, not byte for byte); a float32/float64 comparison in an export check was corrected. The rb99 operating point uses the pre-specified fallback.
8. **Step 6.** No deviation. The machine slept once during training (about 21:04–21:29 IST); the runs were suspended, not restarted, and all 40 finished. We did not check separately that the pause left the weights unchanged.
9. **Steps 7 and 8 (one deviation, `0e49c87`).** The Step 7 and Step 8 evaluators lacked `model.eval()` and cuDNN determinism. This was found when Step 8's anchor failed on its first checkpoint, diagnosed on validation cities only, and fixed with the author's approval before any Step 7 or Step 8 result was saved or read; nothing else changed.
10. **Steps 7 and 8, interruption.** Both were stopped by the author on 2026-10-23 and resumed the next day. Step 8 was re-run from scratch and reproduced the 16 files of the interrupted run byte for byte; in Step 7, 5 completed runs were kept and the interrupted run was retrained from scratch. That retrain was checked only against the interrupted log's last line: at step 1925 its loss (−0.2677) and validation score (0.1623) match to the four decimals the log keeps. Weights were not compared, so this is not a bit-identity check.
11. **Error maps.** The selection rule was committed (`afa354e`) before the script first ran; no deviation.
12. **Revision analyses (2026-11-02).** After a simulated peer review of the manuscript, the exploratory analyses of §3.4 were added. They were chosen after all results were known, read only stored counts and are not pre-registered; no registered analysis or result was changed.
13. **Stage 4B (post-review evaluation).** Registered after the within-B contrast had been seen at B's own `no_o2` threshold and at 0.5; only the clean-threshold, per-city and land-cover counts were unseen. The report generator reuses the column header "Δ (present − absent)" for its descriptive rows, which are other contrasts; the paper's Table 4 labels them correctly.
14. **Line endings (found before Step 9).** Git (`core.autocrlf=true`) had rewritten some committed files with CRLF line endings, so their on-disk hashes no longer matched earlier pre-registrations although their content did. `train_rrd.py`, `verify_prereg_hashes.py` and `smoke_phase6.py` were restored from the commit before Step 9; three files that Steps 9–11 do not use were left with CRLF endings.
15. **Step 9.** E was added after the Stage 3 review named late fusion as an alternative explanation of the RQ1 null; the comparison of E without optical input against the SAR-only U-Net was chosen after Step

9's results were seen and was then tested on new seeds as Step 10's registered primary hypothesis.

16. **Step 10.** The first smoke-test run failed on a JSON type error in the analysis (a numpy integer), fixed before the logged run; the smoke test used only already-reported seed 1–10 runs and printed no metric.
17. **Step 11.** The cloud detector was run on the real scenes before the pre-registration was written, as in Step 5; its masked fractions are stated there and led to rb99 being reported descriptively. The smoke test evaluated seed 1 into a deleted temporary folder without reading any metric. The Las Vegas scene chosen by the "overcast" rule is probably bright desert flagged by the detector; it was kept as the fixed rule selected it.
18. **Steps 9–11, report labels.** The generated reports print labels such as "higher/lower higher"; the paper's labels are derived from the same sign and Holm p-value.

Appendix D: Reproducibility

Each step's pre-registration, report and audit are in `experiments/robust/`: Phases 2–5 (commits `eead015`, `25fe13b`, `5dfaa17`, `cde5434`, `3c5d85a`, `f56ba71`, `d9e29ff`, `5f5be9c`, `3ed2fb7`); Step 6 (pre-registration `5476ffe`, results `2c74a73`); Step 7 (`1ff4251`, `6374275`); Step 8 (`7989248`, `f50d73b`); evaluator deviation `0e49c87`; error maps (`afa354e`, `b49bd87`); Stage 4B (`260c9b2`, `experiments/robust/stage4b/`); Step 9 (`2a4cf09`, `step9/`); Step 10 (`4c721d8`, `step10/`); Step 11 (`50d3d3d`, `realcloud/`). Each pre-registration's hash table can be re-verified with `verify_prereg_hashes.py`. Repeat training was checked twice and gave bit-identical weights. Audit scripts recompute every reported statistic from raw counts: Phase 2, 579 checks; Phase 3, 167; Step 4A-bis, 1338; Step 4B, 432; Step 5S, 826; Step 5, 731; Step 6, 1046; Step 7, 994; Step 8, 235; all with 0 failures (Step 4A has no audit of its own). Stage 4B and Steps 9–11 have no separate audit script: their evaluations enforce anchors (for example, Step 11 reproduces Step 5's stored clean test counts exactly at all 101 thresholds, and Step 10's per-city counts sum to the counts stored at training), and their analyses recompute every rate from stored counts. The exploratory re-analysis is in `experiments/robust/stage4/`, the exploratory contrasts quoted in §4.2–5.5 in `experiments/robust/revision/`, and every number and table cell of this paper is checked against these files by `paper/check_final_paper_numbers.py` and `paper/check_final_paper_tables.py`.

Appendix E: Synthetic and composite cloud details (Steps 4A-bis–5)

Table E1. Step 4A-bis: imperfect masks. For B at 30% cover, each imperfect mask against the oracle mask (Holm over 9); every error type costs F1 (Holm $p < 0.05$, worse than the oracle in 10/10 seeds):

mask	missed cloud	over-masked clear	B F1	Δ vs oracle	95% CI	retained gain
oracle (z30)	0	0	0.516			
dilate2	0.000	0.051	0.513	−0.003	[−0.003, −0.002]	0.99
dilate4	0.000	0.107	0.511	−0.005	[−0.006, −0.005]	0.98
dilate8	0.000	0.214	0.506	−0.010	[−0.012, −0.009]	0.97
erode2	0.116	0.000	0.416	−0.100	[−0.141, −0.060]	0.70
erode4	0.232	0.000	0.329	−0.187	[−0.220, −0.153]	0.43
erode8	0.431	0.000	0.273	−0.243	[−0.272, −0.213]	0.26
miss10	0.100	0.000	0.426	−0.090	[−0.103, −0.077]	0.73
miss25	0.250	0.000	0.348	−0.168	[−0.188, −0.149]	0.49
miss50	0.500	0.000	0.266	−0.250	[−0.276, −0.225]	0.23

Retained gain = (F1_mask − F1_cloud) / (F1_oracle − F1_cloud), computed per seed and averaged, with F1_cloud = 0.191. Over-masking is almost free for B: masking 21% of the clear pixels costs 0.010 F1. Under-

masking is expensive whatever its shape; on missed-cloud pixels, B's false-alarm rate is 0.909–0.962 for every missed-cloud and erode4/8 variant. Over-masking does hurt the full-input model A: its false-alarm rate on over-masked clear pixels rises from ≈ 0.024 to 0.385–0.497. An exploratory partition by the true mask shows that a thin rim of unmasked cloud brings back the Phase 3 spill-over: for B with erode4, the false-alarm rate on clear pixels outside the cloud is 0.073, the same as with no fill (0.075), against 0.045 with the oracle mask.

Table E2. Step 4B: spatial dropout training. C − B test F1 at the frozen clean threshold (Holm over 7):

condition	B	C	Δ (C − B)	95% CI	C > B	Holm p
clean	0.553 ± 0.030	0.564 ± 0.034	+0.011	[−0.024, +0.046]	7/10	1.0
z10	0.539 ± 0.028	0.553 ± 0.033	+0.014	[−0.019, +0.048]	7/10	1.0
z30	0.516 ± 0.028	0.529 ± 0.032	+0.013	[−0.020, +0.047]	6/10	1.0
z60	0.499 ± 0.030	0.508 ± 0.028	+0.009	[−0.024, +0.041]	6/10	1.0
c30, erode4	0.329 ± 0.029	0.312 ± 0.034	−0.018	[−0.050, +0.015]	2/10	1.0
c30, dilate4	0.511 ± 0.028	0.522 ± 0.030	+0.011	[−0.021, +0.043]	6/10	1.0
c30, miss25	0.348 ± 0.007	0.341 ± 0.006	−0.007	[−0.015, +0.001]	3/10	0.55

All 7 primary comparisons are "no detectable difference", and C shows no clean regression. C halves B's whole-image dropout, so the contrast mixes spatial masks with less whole-image dropout; descriptively, C is lower than B on `no_opt` (−0.028) and `o2_copy` (−0.039), and C still calls almost every unmasked cloud pixel change (false-alarm rate under the cloud with no fill 0.994, against 0.961 for B).

Table E3. Step 5S: dilated masks. B at 30% cover, test F1 with each under-masking mask P and with P dilated by r* = 16 px (Holm over 6), and the remaining difference from the oracle mask after dilation (secondary):

mask	B F1, P	B F1, P dilated by r*	Δ	95% CI	Δ > 0	Holm p	Δ vs oracle at r*
erode2	0.416 ± 0.050	0.500 ± 0.027	+0.085	[+0.044, +0.125]	10/10	0.0041	−0.016
erode4	0.329 ± 0.029	0.500 ± 0.027	+0.171	[+0.138, +0.204]	10/10	4.3e-06	−0.016
erode8	0.273 ± 0.018	0.468 ± 0.020	+0.194	[+0.172, +0.216]	10/10	6.0e-08	−0.048
miss10	0.426 ± 0.013	0.424 ± 0.015	−0.002	[−0.005, +0.001]	4/10	0.2382	−0.092
miss25	0.348 ± 0.007	0.348 ± 0.008	+0.000	[−0.001, +0.002]	5/10	0.7405	−0.168
miss50	0.266 ± 0.010	0.267 ± 0.011	+0.002	[+0.001, +0.003]	9/10	0.0261	−0.249

Dilation is better for all three eroded masks, with every seed improving, and afterwards they are within 0.048 F1 of the oracle. For missed whole clouds it makes no practical difference: miss50 meets the label "dilation better" with an effect of 0.002 F1 that the Holm-corrected Wilcoxon test does not support (p 0.059). Dilation can only reach a missed cloud that lies near a detected one. The validation score was still rising at r* = 16 px, the edge of the grid.

Table E4. Step 5: composite clouds and a real detector. These composites are not real cloudy acquisitions: thin cloud carries the foreign surface from its source scene, there are no matching shadows, and the atmosphere outside the cloud is unchanged. Primary family: B, test F1 at the frozen clean threshold, Holm over 9.

contrast	Δ	95% CI	Δ > 0	Holm p	label
c10: rb95 – bal	−0.009	[−0.019, +0.001]	3/10	0.0672	no detectable difference
c10: rb95 – oracle	−0.055	[−0.065, −0.045]	0/10	4.4e-06	rb95 worse
c10: rb95 – cloud	+0.036	[+0.017, +0.055]	10/10	0.0056	rb95 better
c30: rb95 – bal	+0.023	[+0.008, +0.037]	9/10	0.0119	rb95 better
c30: rb95 – oracle	−0.053	[−0.067, −0.039]	0/10	6.7e-05	rb95 worse
c30: rb95 – cloud	+0.150	[+0.125, +0.175]	10/10	2.1e-06	rb95 better
c60: rb95 – bal	+0.057	[+0.037, +0.078]	10/10	5.6e-04	rb95 better
c60: rb95 – oracle	−0.020	[−0.024, −0.016]	0/10	6.8e-06	rb95 worse
c60: rb95 – cloud	+0.257	[+0.223, +0.291]	10/10	3.4e-07	rb95 better

Detector masks on the test composites and B's mean test F1 (descriptive). Cloud recall and the fraction of clear pixels masked are computed against the true mask with no change model; rb99 is the flagged fallback mask.

level	rb95 cloud recall	rb95 clear masked	bal cloud recall	bal clear masked	B cloud	B bal	B rb95	B rb99	B oracle	rb95 retained gain
c10	0.892	0.596	0.532	0.069	0.447	0.492	0.483	0.478	0.538	0.35 ± 0.21
c30	0.837	0.615	0.555	0.087	0.315	0.442	0.465	0.478	0.518	0.74 ± 0.06
c60	0.963	0.673	0.747	0.094	0.217	0.417	0.474	0.478	0.494	0.93 ± 0.02

The recall-biased mask beats the most accurate one at 30% and 60% cover and beats no fill at every level, but stays below the oracle mask. It retains 0.740 of the oracle gain at 30% cover, just below the pre-specified 0.75, so no pre-specified decision branch applies exactly. Three results qualify the gain: the detector misses more cloud on test than on validation (11–16% of test cloud pixels reach the model at 10% and 30% cover); rb95 reaches its recall by masking 0.596–0.673 of the clear pixels and costs B 0.054 F1 on clean images; and rb99, which masks almost the whole post-change image, gives 0.478 at every level, above rb95 at 30% cover. The gain over bal is therefore consistent with B falling back on SAR and the pre-change image, not evidence that the detector locates cloud well. For A, rb95 is worse than bal at every level (−0.148, −0.083, −0.032; Holm $p < 0.05$).

Table E5. SAR-only fallback rule on composite clouds (exploratory). A simpler operational rule is to use A when the post-change optical image is available and clear, and the SAR-only U-Net otherwise. The SAR-only score (0.418 ± 0.043) does not depend on the optical image, so it is the same at every cloud level. Mean difference B − SAR-only in test F1 with its 95% CI over seeds, B at its frozen clean threshold on Step 5's composite clouds, by cloud mask:

cloud cover	no mask	bal	rb95	rb99	oracle
c10	+0.029 [−0.004, +0.062]	+0.074 [+0.041, +0.108]	+0.065 [+0.032, +0.098]	+0.060 [+0.026, +0.094]	+0.120 [+0.086, +0.154]
c30	−0.103 [−0.137, −0.070]	+0.024 [−0.010, +0.058]	+0.047 [+0.014, +0.079]	+0.059 [+0.026, +0.093]	+0.099 [+0.067, +0.132]
c60	−0.202 [−0.236, −0.167]	−0.001 [−0.036, +0.033]	+0.056 [+0.023, +0.088]	+0.060 [+0.026, +0.094]	+0.076 [+0.043, +0.109]

When the post-change image is available and clear, the rule uses A: B is not detectably different in F1 (−0.003) but has lower precision (−0.101, 95% CI [−0.161, −0.041]), higher recall (+0.120) and a higher false-alarm rate (+0.018). When the image is missing, the rule uses SAR-only, which B does not detectably beat at its registered threshold (+0.013). On composite clouds without a mask, B is below the SAR-only fallback at 30% and 60% cover; with the recall-biased detector mask rb95 it is above it at every level, and with rb99, which discards almost the whole post-change image, by +0.059 to +0.060. Because rb99 nearly removes the image, much of B's advantage here is the clean-threshold margin of §4.2 rather than information from the masked image. These comparisons use composites, not real cloudy acquisitions, and were made after the results were known.